MHF4U Advanced Functions - Unit 1 Practice Test, Version 2

Worked key and thinking guide

50 marks. The category labels on the student paper identify what each part assesses. Award equivalent correct reasoning. The graph questions require labelled key points and correct open or closed endpoints.

1. Piecewise functions [9]

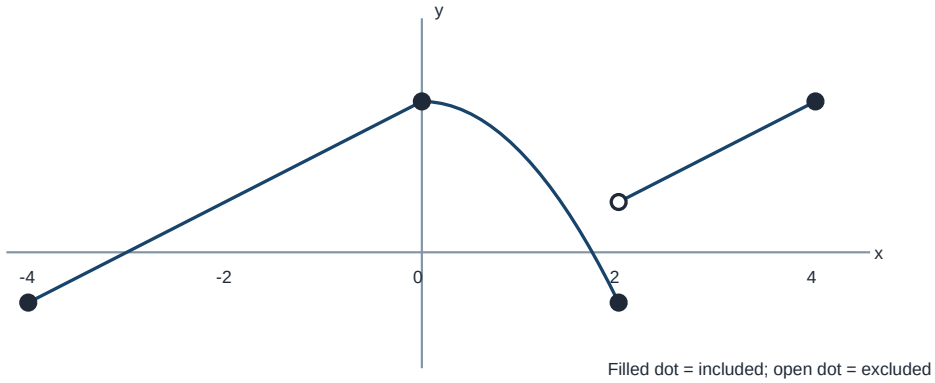
a. Draw the line y=x+3 from filled `(-4,-1)` toward open `(0,3)`; draw y=3-x^2 from filled `(0,3)` through `(1,2)` to filled `(2,-1)`; draw y=x-1 for 2<x<=4, with open `(2,1)` and filled `(4,3)`. [4]

b. Domain `[-4,4]`; range `[-1,3]`. [2]

c. The first branch increases from x=-4 to x=0, and the third branch increases for 2<x<=4. Accept `[-4,0]` and `(2,4]` as branch intervals. [1]

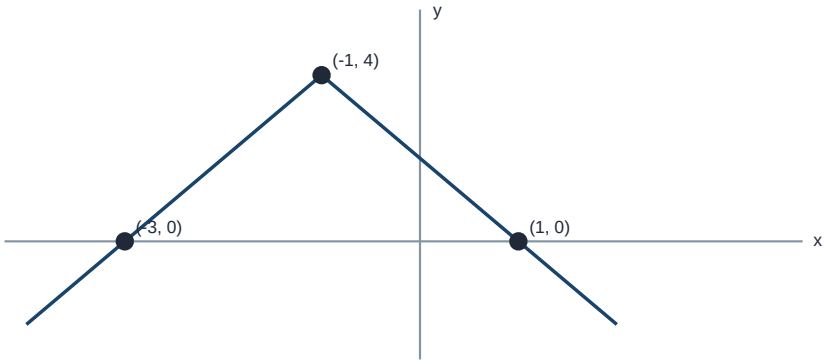
d. No. A horizontal line such as y=2 meets the graph at more than one point: P(-1)=2, P(1)=2, and P(3)=2. After swapping coordinates, input 2 would have three outputs. [2]

Thinking move: To decide whether an inverse is a function, test horizontal lines on the original graph; a repeated output becomes a repeated input in the inverse.



2. Absolute value graph [6]

The vertex is `(-1,4)` and the V opens down. Solve `-2|x+1|+4=0`, so `|x+1|=2` and x-intercepts are `(-3,0)` and `(1,0)`. Label these on the sketch. [3] Domain `(-infinity,infinity)`; range `(-infinity,4]`. [2] The negative sign reflects the parent V in the x-axis, and magnitude 2 stretches it vertically by factor 2. [1]



3. Transforming a function [5]

a. Match the original input 4: `-2(x-1)=4`, hence x=-1. The new output is 3(2)-1=5, so the point is `(-1,5)`. [2]

b. The square-root input must satisfy `-2(x-1)>=0`, so domain `(-infinity,1]`. Since f is nonnegative, g>=-1; range `[-1,infinity)`. [2]

c. The horizontal transformation reflects the graph in the y-axis, compresses it by factor 1/2, and shifts it right 1. [1]

Thinking move: For a point on a transformed graph, set the whole inside expression equal to the old x-coordinate; do not multiply the old x-coordinate by the inside coefficient.

4. Inverse relations [5]

a. Original q has domain all reals and range `[-1,infinity)`. The inverse relation therefore has domain `[-1,infinity)` and range `(-infinity,infinity)`. [2]

b. The parabola fails the horizontal-line test: for example `q(-1)=0=q(-3)`, so the inverse would send 0 to two outputs. [1]

c. One valid restriction is `x>=-2`; swap x and y in `y=(x+2)^2-1`, then take the positive root to obtain `q^(-1)(x)=-2+sqrt(x+1)`, domain `x>=-1`. Accept the alternative restriction `x<=-2` and inverse `-2-sqrt(x+1)`. [2]

Thinking move: Restrict to one side of the vertex before solving the inverse. The restriction decides which square-root branch is valid.

5. Observation wheel diagram [7]

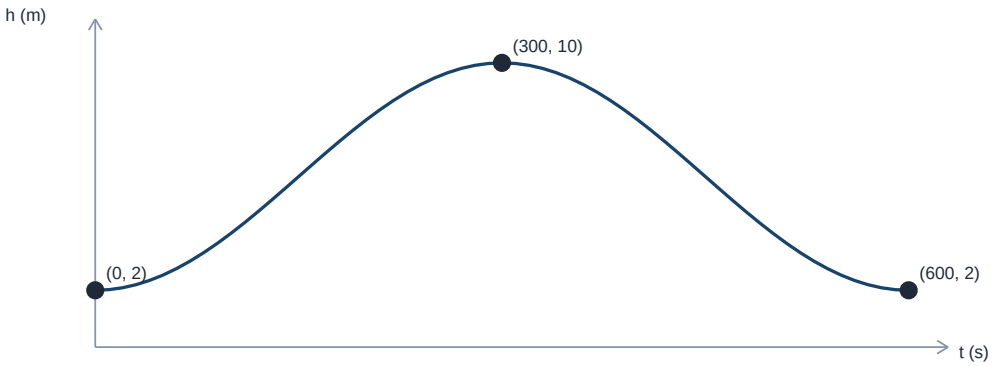
a. Centre height `2+4=6 m`; greatest seat height `6+4=10 m`. [2]

b. Domain `[0,600]` seconds; range `[2,10]` metres. [2]

c. Twelve metres adds the radius to the already maximum height, or treats the centre as 8 m. The centre is 6 m, so the top is 10 m. [1]

d. Draw one smooth cycle starting at `(0,2)`, rising to `(300,10)`, and falling to `(600,2)`. Label the axes time (s) and height (m). [2]

Thinking move: Read the bottom clearance and radius separately. Minimum = clearance; centre = minimum + radius; maximum = centre + radius. Half a rotation takes half the time.



6. Average and instantaneous rates [8]

a. `H(0)=1` and `H(2)=33`; average rate `(33-1)/(2-0)=16 m/s`. [2]

b. `H(1.9)=32.16` and `H(2.1)=33.76`; centred secant estimate `(33.76-32.16)/(2.1-1.9)=8 m/s`. [3]

c. Complete the square: `H(t)=-4(t-3)^2+37`. The maximum is `37 m` at `t=3 s`. [2]

d. The first rate averages motion over two seconds, during which the ball's upward speed changes. The second estimates the rate only near `t=2`; the two intervals describe different aspects of the same motion. [1]

Thinking move: A secant slope uses two separated times. A tangent estimate uses times very close to the instant of interest. Always attach height-per-time units.

7. Graph properties and interval notation [5]

a. `r(-x)=(-x)^3-4(-x)=-x^3+4x=-(x^3-4x)=-r(x)`, so `r` is odd. [2]

b. Origin symmetry gives `(-3,-15)`. [1]

c. `|x-1|<=3` means `-3<=x-1<=3`, so `-2<=x<=4`, or `[-2,4]`. [2]

8. Secant and tangent diagram [5]

a. Line `AB` is the secant and represents the average rate: `(5-1)/(3-1)=2` units of y per unit of `x`. [2]

b. The horizontal line through `A` is tangent at `A` and represents the instantaneous rate there; slope `0`. [2]

c. The secant measures the whole interval from 1 to 3, where the curve rises. The tangent describes the curve's direction only at the minimum `A`, where it is momentarily flat. [1]

Thinking move: Identify the interval first: average rate is a secant. Identify the single x-value next: instantaneous rate is a tangent.